

EXPLORATORY DATA ANALYSIS (EDA) REPORT

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Project: Project 2 – Exploratory Data Analysis (EDA)

Tool Used: Python (Pandas, Matplotlib)

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on the cleaned dataset to identify patterns, trends, distributions, and outliers. The analysis helps in understanding customer behavior and generating meaningful business insights for decision-making.

Dataset Information

- Dataset Used: Cleaned_Dataset.xlsx
 - Total Records: 1200
 - Total Columns: 14
 - Missing Values: 0
 - Duplicate Records: 0
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Exploratory Data Analysis Performed

1. Dataset Overview

The dataset was explored using:

- Shape
 - Columns
 - Data Types
 - Dataset Information
 - Descriptive Statistics
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2. Descriptive Statistics

Basic statistical analysis was performed on numerical columns to calculate:

- Count
- Mean
- Median
- Standard Deviation

- Minimum Value
 - Maximum Value
 - Quartiles
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3. Product Analysis

The distribution of products was analyzed to identify the most frequently ordered products.

Visualization Used:

- Bar Chart
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4. Payment Method Analysis

Customer payment preferences were analyzed.

Visualization Used:

- Pie Chart
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5. Order Status Analysis

The distribution of order statuses was examined to understand the order completion process.

Visualization Used:

- Bar Chart
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6. Referral Source Analysis

Customer acquisition sources were analyzed.

Visualization Used:

- Bar Chart
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7. Coupon Usage Analysis

The usage of coupons was analyzed to understand promotional effectiveness.

Visualization Used:

- Pie Chart
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8. Items in Cart Analysis

The number of items purchased per order was analyzed.

Visualization Used:

- Histogram
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9. Sales Analysis

The following sales metrics were calculated:

- Total Revenue
- Average Order Value
- Highest Order Value
- Lowest Order Value

Monthly sales trends were also analyzed.

Visualization Used:

- Line Chart
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10. Correlation Analysis

Correlation among numerical variables was calculated to understand relationships between:

- Quantity
 - Unit Price
 - Items in Cart
 - Total Price
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11. Outlier Detection

Outliers were identified using a Box Plot on the Total Price column to detect unusually high or low order values.

Visualization Used:

- Box Plot

Business Insights

1. The dataset was clean and free from missing values after preprocessing.
 2. No duplicate records were found, ensuring data reliability.
 3. Product demand varied across different products.
 4. Customers preferred specific payment methods over others.
 5. Coupon usage was lower compared to non-coupon transactions.
 6. Monthly sales showed noticeable variations.
 7. A small number of high-value orders were identified as potential outliers.
 8. Referral sources contributed differently to customer acquisition.
 9. Average order value provided insights into customer purchasing behavior.
 10. The dataset is suitable for dashboard creation and advanced predictive analytics.
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Conclusion

Exploratory Data Analysis (EDA) was successfully performed on the cleaned dataset. Statistical analysis, trend identification, correlation analysis, and outlier detection provided meaningful insights into customer purchasing behavior. The project demonstrates practical knowledge of Python, Pandas, Matplotlib, and analytical thinking for solving real-world business problems.

Tools & Technologies Used

- Python
 - Pandas
 - Matplotlib
 - Jupyter Notebook
 - Microsoft Excel
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Project Outcome

The project successfully transformed raw business data into meaningful analytical insights. The cleaned dataset and EDA findings can be used for reporting, dashboard development, and future predictive analytics projects.